

Decision Tree Model and Business Interpretation

FreshBasket Subscription Grocery Delivery: Customer Churn Prediction

Prepared for: Assignment 4 - Decision Tree Classification and Business Interpretation

Course: MBAI 5310G - AI Programming

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Model Used: Decision Tree Classifier

Target Variable: Churned

Positive Class: Yes

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1. Executive Summary

This report presents the Assignment 4 Decision Tree classification project for FreshBasket, a subscription-based grocery delivery service. The purpose of the model is to predict whether a customer is likely to churn so that the business can take timely retention action before losing recurring revenue.

The original dataset contained 425 rows and 16 columns. After removing 5 exact duplicate rows, the final cleaned dataset contained 420 rows and 16 columns. The target variable is Churned, where Yes indicates that the customer cancelled or did not renew the subscription.

The Decision Tree model achieved a testing accuracy of 72.62%. For the positive class, Yes, the model achieved precision of 50.00%, recall of 60.87%, and F1-score of 54.90%. The training accuracy was 75.60% and the testing accuracy was 72.62%, producing an accuracy gap of 2.98%.

From a business perspective, recall for the churned class is especially important because a false negative means the company misses a customer who is likely to leave. The model should therefore be used as a decision-support tool for prioritizing customer retention actions, not as an automatic decision system.

2. Introduction and Assignment Objective

The objective of this assignment is to build and interpret a Decision Tree classification model based on a business plan. The assignment requires both technical implementation and business interpretation. Therefore, this report covers the complete workflow from business understanding and data preparation to model evaluation, feature importance, limitations, and responsible use.

The problem is a supervised binary classification problem because the target variable has two possible outcomes: Yes and No. The model uses customer profile, subscription, service-quality, spending, and engagement variables to predict customer churn risk.

3. Business Context and Machine Learning Decision

FreshBasket is a subscription-based grocery delivery service operating in the Greater Toronto Area. It offers weekly and monthly grocery delivery plans for busy professionals, families, and customers who prefer online ordering and convenient delivery. The company earns revenue through recurring subscriptions and regular grocery purchases.

The main business problem is customer churn. When customers cancel or do not renew, FreshBasket loses recurring revenue and may need to spend more on acquiring replacement customers. Machine learning can help the company identify customers who may need retention attention before they leave.

The machine learning decision supported by the model is: Which customers should FreshBasket proactively contact with retention support before they churn?

Table 1: Business question and machine learning decision

Business Question	Machine Learning Decision
Which customers are likely to cancel or not renew?	Predict Churned = Yes or Churned = No.
Which factors explain churn risk?	Use Decision Tree feature importance to identify important churn-related signals.
Where should the business focus retention actions?	Prioritize high-risk customers and improve the strongest churn-related factors.

4. Dataset Overview

The dataset used for this project is `freshbasket_customer_churn_dataset.xlsx`. Each row represents one FreshBasket customer. The dataset includes demographic, region, subscription, payment, delivery, spending, service-quality, discount, engagement, satisfaction, and churn outcome variables.

Table 2: Dataset and project summary

Item	Value
Original dataset shape	425 rows x 16 columns
Duplicate rows before cleaning	5
Duplicate rows after cleaning	0
Final cleaned dataset shape	420 rows x 16 columns
Target variable	Churned
Positive class	Yes = customer churned
Problem type	Supervised binary classification
Model used	Decision Tree Classifier

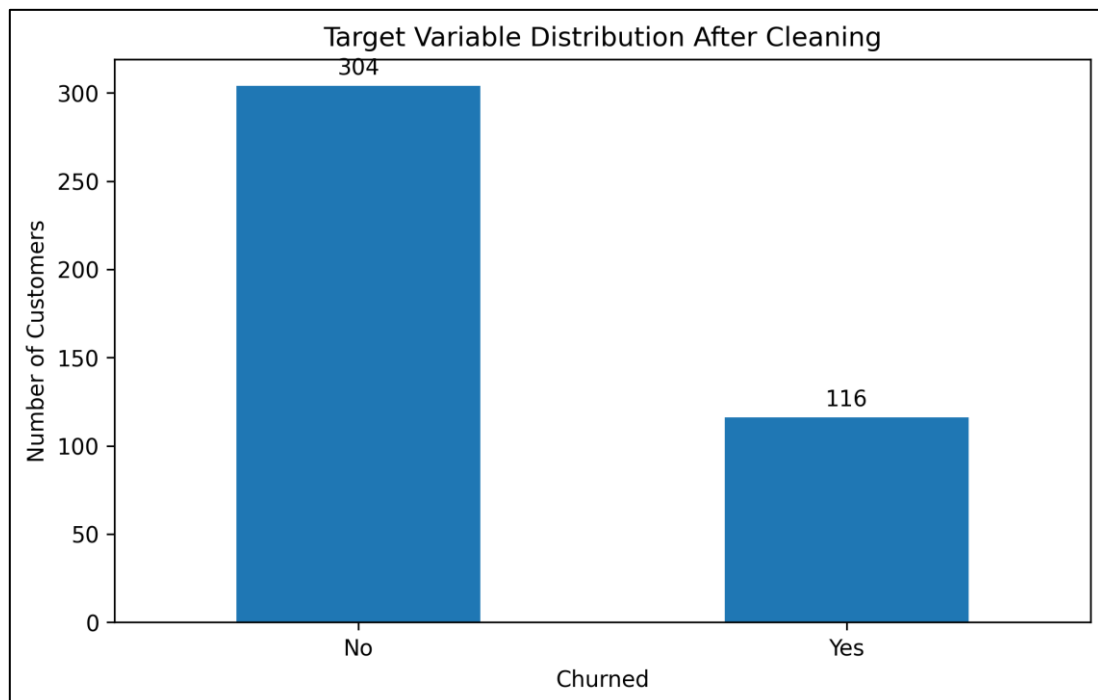


Figure 1: Target variable distribution after cleaning

After duplicate removal, the cleaned dataset contains 304 customers in the No class and 116 customers in the Yes class. The No class is larger than the Yes class, so accuracy should be interpreted together with precision, recall, F1-score, and the confusion matrix.

5. Target Variable and Input Features

The target variable is Churned. The value Yes means the customer churned, while No means the customer did not churn. The model uses the remaining business variables as input features after removing Customer_ID, because it is an identifier rather than a behaviour variable.

Table 3: Dataset columns and business meaning

Column	Type	Business Meaning
Customer_ID	Identifier	Unique customer code; removed before model training.
Age_Group	Categorical	Customer age segment.
Region	Categorical	Customer service region in the Greater Toronto Area.
Subscription_Type	Categorical	Selected subscription plan.
Payment_Method	Categorical	Main payment method used by the customer.
Delivery_Window	Categorical	Preferred delivery time window.
Membership_Length_Months	Numerical	Number of months since the customer joined.
Monthly_Spend	Numerical	Customer monthly spend in Canadian dollars.
Average_Order_Value	Numerical	Average dollar value of customer orders.
Orders_Last_3_Months	Numerical	Number of orders placed in the last three months.
Delivery_Issues_Last_3_Months	Numerical	Number of reported delivery issues in the last three months.
Customer_Service_Tickets	Numerical	Number of support tickets submitted by the customer.
Discount_Used	Categorical	Whether the customer used a discount recently.
App_Engagement_Score	Numerical	Score from 0 to 100 showing app usage and interaction.
Satisfaction_Score	Numerical	Customer satisfaction rating from 1 to 5.
Churned	Target	Whether the customer cancelled or did not renew.

Table 4: Numerical and categorical feature groups

Feature Type	Number of Features	Columns
Numerical features	8	Membership_Length_Months, Monthly_Spend, Average_Order_Value, Orders_Last_3_Months, Delivery_Issues_Last_3_Months, Customer_Service_Tickets, App_Engagement_Score, Satisfaction_Score
Categorical features	6	Age_Group, Region, Subscription_Type, Payment_Method, Delivery_Window, Discount_Used

6. Data Preparation and Data Quality Review

Data preparation was completed before model training to improve data quality and make the dataset suitable for machine learning. The inspection included checking the dataset shape, column names, data types, missing values, duplicate rows, target distribution, and feature groups.

Table 5: Target variable distribution before and after cleaning

Stage	No	Yes	Total
Before duplicate removal	307	118	425
After duplicate removal	304	116	420

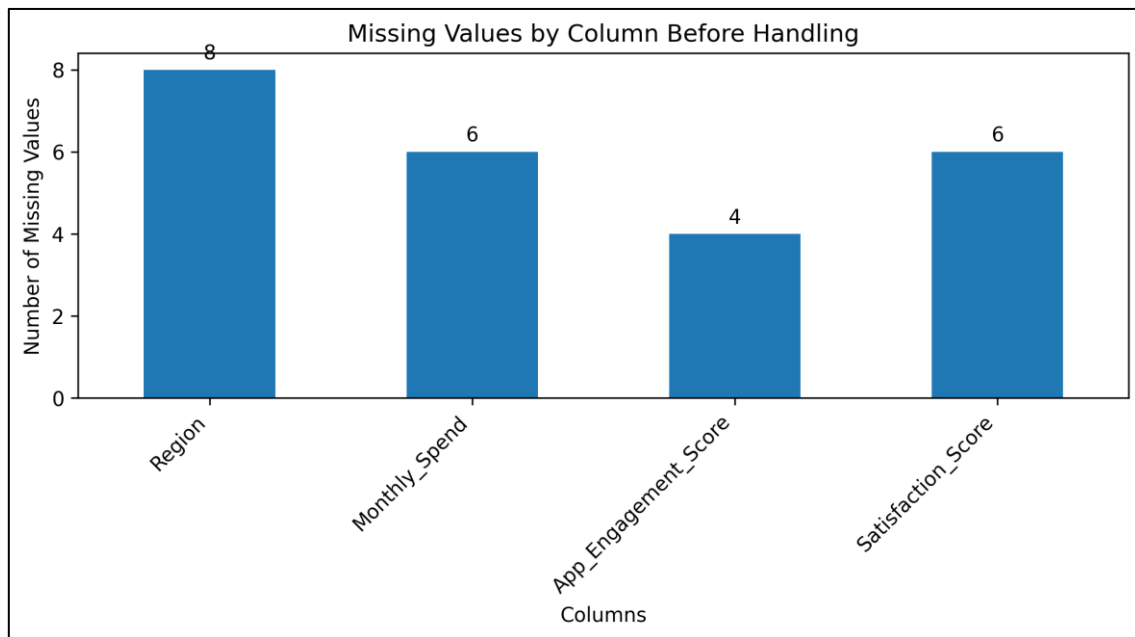


Figure 2: Missing values by column before handling

Table 6: Missing value inspection and handling plan

Column	Data Type	Missing Values	Planned Method	Handling Location
Region	Categorical	8	Most frequent value	Inside preprocessing pipeline
Monthly_Spend	Numerical	6	Median	Inside preprocessing pipeline
App_Engagement Score	Numerical	4	Median	Inside preprocessing pipeline
Satisfaction Score	Numerical	6	Median	Inside preprocessing pipeline

Missing values were not filled directly in the full dataset before the train-test split. Instead, missing values were handled later inside the preprocessing pipeline. Numerical missing values were handled using median imputation, while categorical missing values were handled using the most frequent value. This helps avoid data leakage because the preprocessing transformer is fitted on the training data and then applied to the testing data.

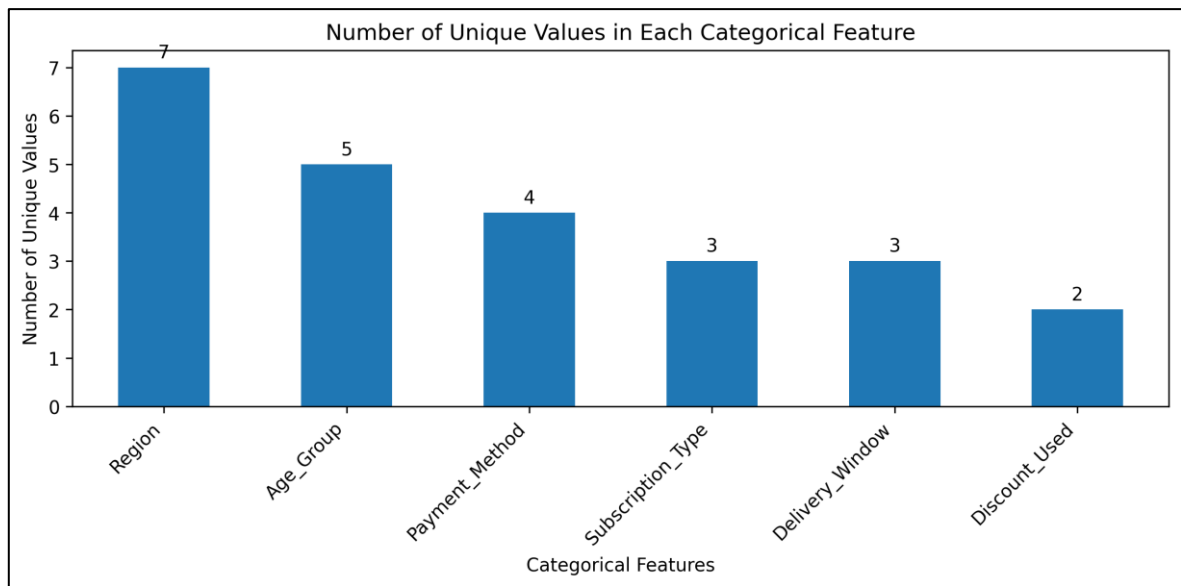


Figure 3: Number of unique values in each categorical feature

7. Feature/Target Separation and Data Splitting

After cleaning duplicate rows, the dataset was separated into the feature matrix X and target vector y . The target variable Churned was separated from the inputs, and Customer_ID was removed because it is an identifier.

Table 7: Feature matrix and target vector

Object	Shape	Meaning
X	420 rows x 14 feature columns	Input variables used by the model
y	420 values	Target variable: Churned

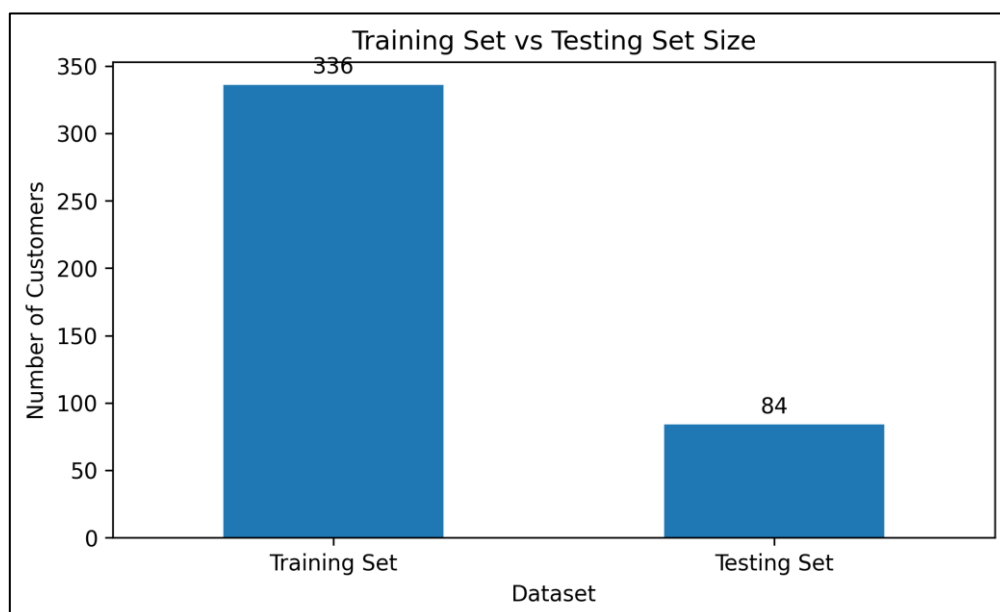


Figure 4: Training set vs testing set size

Table 8: Train-test split summary

Dataset	Shape
X_train	(336, 14)
X_test	(84, 14)
y_train	(336,)
y_test	(84,)

Table 9: Target class distribution in training and testing sets

Target Class	Training Set	Testing Set
No	243	61
Yes	93	23

The dataset was split using an 80/20 split with random_state=42 and stratify=y. Stratification helps maintain a similar target class distribution in the training and testing datasets.

8. Preprocessing and Decision Tree Modelling

Machine learning models require numerical input. Therefore, preprocessing was applied to both numerical and categorical variables. Numerical missing values were handled using median imputation. Categorical missing values were handled using the most frequent value, and categorical variables were converted into numerical columns using one-hot encoding.

Table 10: Preprocessing approach

Feature Group	Columns	Transformation
Numerical features	Membership_Length_Months, Monthly_Spend, Average_Order_Value, Orders_Last_3_Months, Delivery_Issues_Last_3_Months, Customer_Service_Tickets, App_Engagement_Score, Satisfaction_Score	SimpleImputer(strategy="median"); no scaling required for a Decision Tree
Categorical features	Age_Group, Region, Subscription_Type, Payment_Method, Delivery_Window, Discount_Used	SimpleImputer(strategy="most_frequent") and OneHotEncoder(handle_unknown="ignore")

Table 11: Decision Tree model configuration

Model Item	Value
Algorithm	DecisionTreeClassifier
Maximum depth	7
Minimum samples per leaf	15
Class weight	balanced
Random state	42
Training input shape after preprocessing	(336, 32)
Testing input shape after preprocessing	(84, 32)
Number of processed features	32

The tree was controlled using `max_depth=7` and `min_samples_leaf=15` to reduce overfitting and keep the tree interpretable. The `class_weight="balanced"` setting was used because the churned class is smaller than the non-churned class and recall for churned customers is important for the business.

9. Model Evaluation Results

The trained Decision Tree model was evaluated on the testing data. Predictions were compared with actual values using accuracy, precision, recall, F1-score, a classification report, and a confusion matrix.

Table 12: Sample actual and predicted values

Actual	Predicted
No	No
No	No
No	Yes
No	No
No	No
Yes	Yes
Yes	No
No	No
No	No
No	No

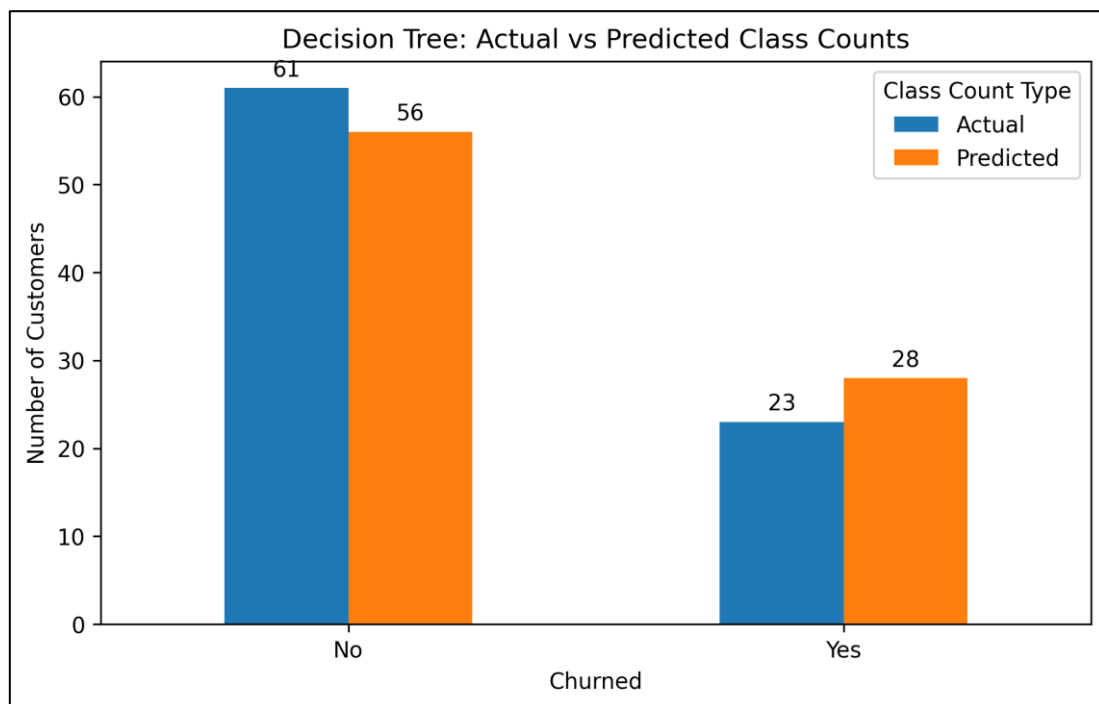


Figure 5: Actual vs predicted class counts

Table 13: Decision Tree evaluation metrics

Metric	Score	Percentage
Accuracy	0.7262	72.62%
Precision for Yes	0.5000	50.00%
Recall for Yes	0.6087	60.87%
F1-score for Yes	0.5490	54.90%

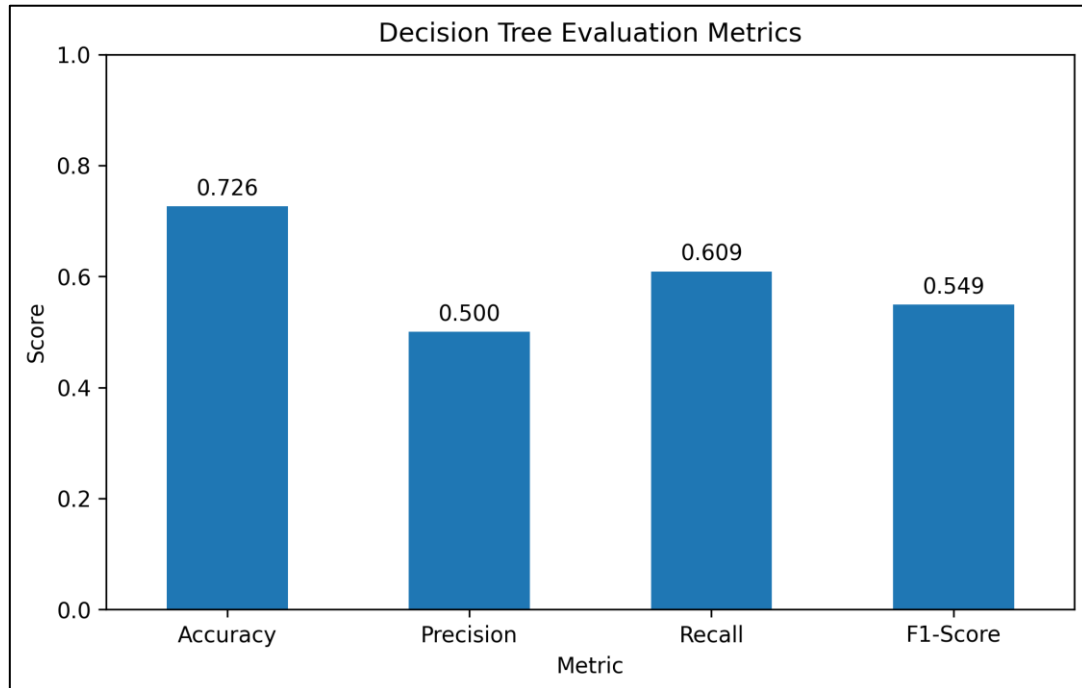


Figure 6: Decision Tree evaluation metrics

Table 14: Classification report

Class/Metric	Precision	Recall	F1-score	Support
No	0.8393	0.7705	0.8034	61
Yes	0.5000	0.6087	0.5490	23
accuracy			0.7262	84
macro avg	0.6696	0.6896	0.6762	84
weighted avg	0.7464	0.7262	0.7338	84

The model correctly classified 72.62% of testing records. The Yes-class precision of 50.00% means that when the model predicted a customer as churned, it was correct about 50% of the time. The Yes-class recall of 60.87% means that the model identified about 61% of the actual churned customers.

For this business problem, recall for the Yes class is especially important because missed churned customers may not receive retention support before leaving. Accuracy alone is not sufficient because the dataset contains more No cases than Yes cases.

10. Confusion Matrix and Business Error Analysis

Table 15: Confusion matrix values

Actual / Predicted	Predicted No	Predicted Yes
Actual No	47	14
Actual Yes	9	14

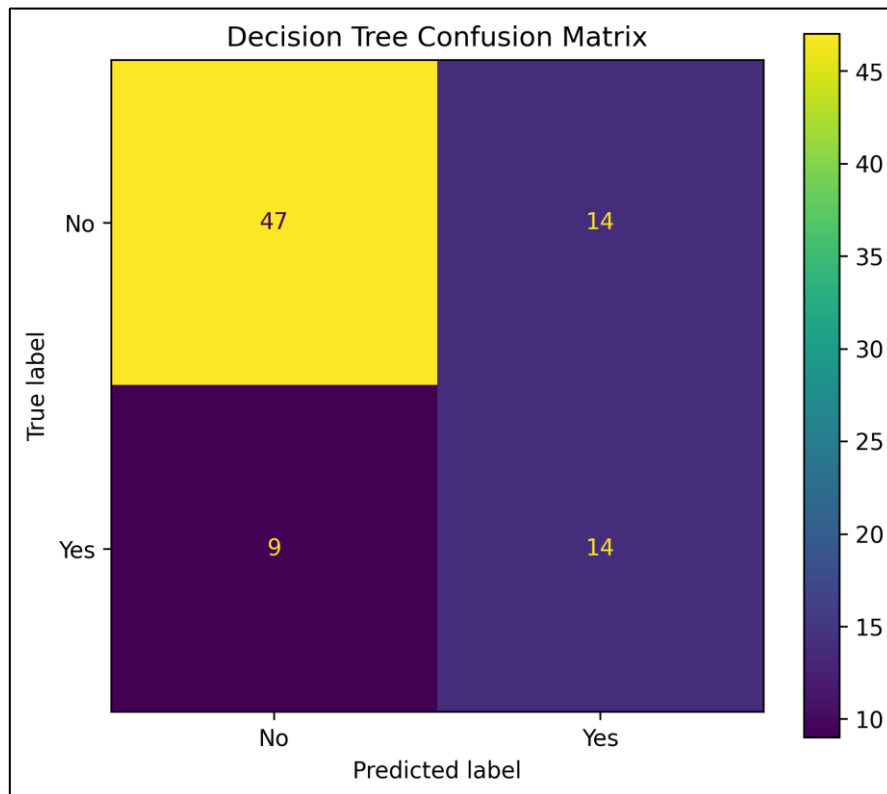


Figure 7: Decision Tree confusion matrix

Table 16: Confusion matrix business meaning

Item	Value	Business Meaning
True Negative	47	Customers correctly predicted as not churned.
False Positive	14	Customers incorrectly predicted as churned; may receive unnecessary retention contact.
False Negative	9	Churned customers incorrectly predicted as not churned; may be missed by retention efforts.
True Positive	14	Churned customers correctly identified for possible retention support.

The model produced 14 true positives and 47 true negatives. It also produced 14 false positives and 9 false negatives. In the FreshBasket context, false negatives are especially important because they represent actual churned customers who may not receive timely support. False positives still matter because they may create unnecessary support work or discount costs.

11. Feature Importance and Business Interpretation

Decision Trees provide feature importance values that indicate which variables contributed most strongly to the model splits. These values help translate model behaviour into practical business insight.

Table 17: Top 10 feature importance values

Feature	Importance	Percentage
Satisfaction_Score	0.359462	35.95%
Monthly_Spend	0.242482	24.25%
Orders_Last_3_Months	0.157547	15.75%
App_Engagement_Score	0.135484	13.55%
Average_Order_Value	0.046943	4.69%
Age_Group_25-34	0.042778	4.28%
Discount_Used_Yes	0.015305	1.53%
Membership_Length_Months	0.000000	0.00%
Subscription_Type_Premium	0.000000	0.00%
Age_Group_18-24	0.000000	0.00%

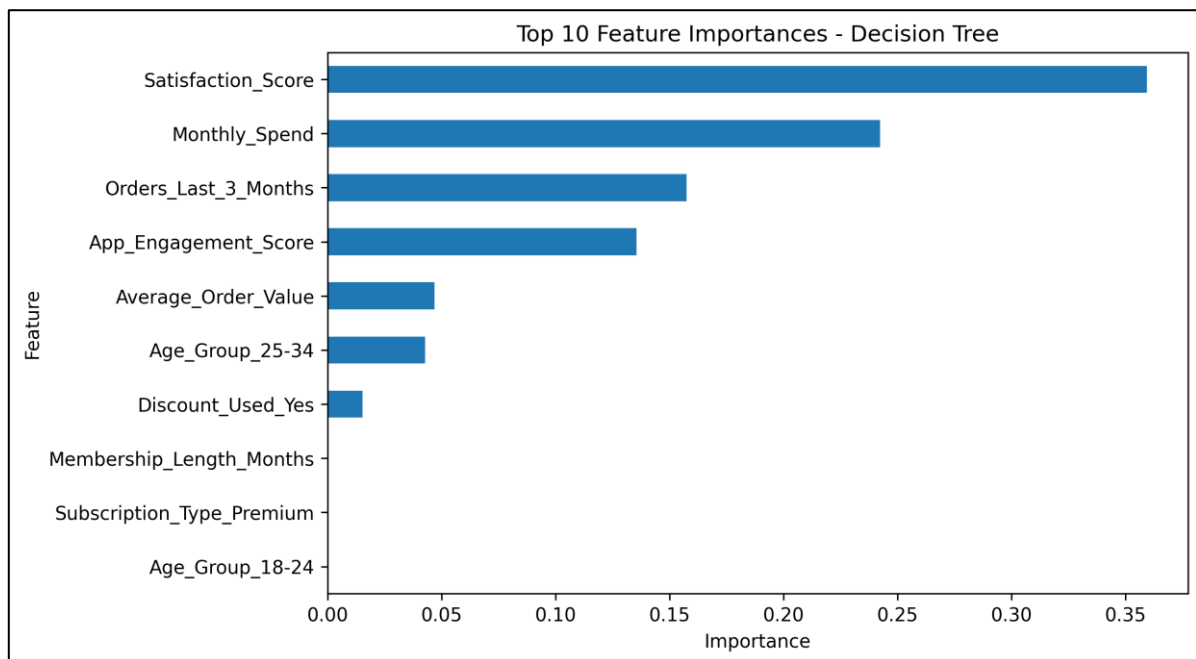


Figure 8: Top 10 feature importances - Decision Tree

The most important features were connected to satisfaction, monthly spend, recent order activity, app engagement, average order value, age group, discount use, membership length, and subscription type. This suggests that churn risk is influenced by customer experience, purchasing behaviour, engagement level, and subscription-related signals.

Feature importance values show which variables were useful for the model's splits, but they do not prove that these variables directly cause churn. Therefore, the results should be interpreted as decision-support insights rather than causal conclusions.

12. Overfitting and Generalization

Table 18: Training and testing accuracy comparison

Evaluation Set	Accuracy	Percentage
Training Set	0.7560	75.60%
Testing Set	0.7262	72.62%
Difference	0.0298	2.98%

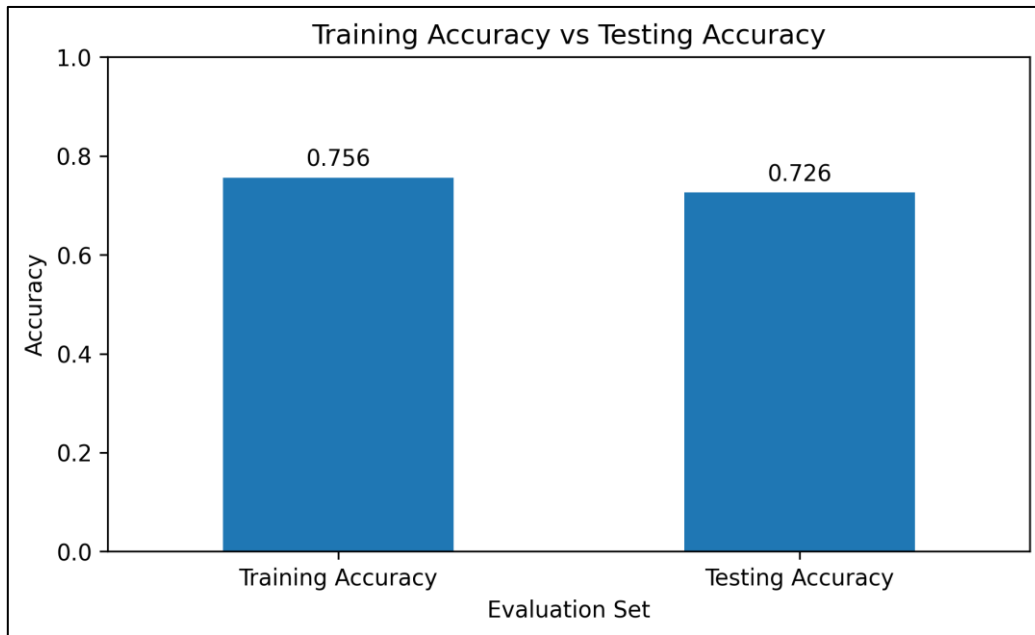


Figure 9: Training accuracy vs testing accuracy

The training accuracy was 75.60%, while the testing accuracy was 72.62%. The difference was approximately 2.98%. This gap is relatively small, so the selected Decision Tree does not show strong signs of overfitting. The `max_depth`, `min_samples_leaf`, and `class_weight` settings appear to support a reasonable balance between business usefulness and generalization.

13. Business Recommendations

The model should be used as a screening tool to support customer retention planning. FreshBasket can use the predictions and important features to prioritize outreach, but human review should guide final actions.

Table 19: Business recommendations linked to important features

Signal / Feature	Business Interpretation	Recommended Action
Low satisfaction score	Customer experience problems may increase churn risk.	Contact the customer with a service recovery message or support follow-up.
Low recent order activity	Fewer recent orders may signal declining usage or lower interest in the service.	Send personalized reminders, product suggestions, or plan recommendations to encourage continued ordering.

Signal / Feature	Business Interpretation	Recommended Action
Low app engagement	Lower app use may show weakening relationship with the service.	Send app reminders, easier ordering tips, or personalized product recommendations.
High service tickets	Repeated support needs can indicate friction or dissatisfaction.	Review unresolved tickets and prioritize customer support follow-up.
Low or changing spend patterns	Lower spending may signal lower usage or declining interest.	Recommend a better subscription plan or personalized grocery bundle.
Discount use	Some customers may respond to targeted incentives.	Use discounts carefully, focusing on high-risk customers where retention value justifies the cost.

A practical implementation approach would be to review customers predicted as Yes each week, combine the model output with staff knowledge, and assign the most appropriate retention action. The model should not automatically trigger discounts for every predicted churned customer; the business should match the intervention to the reason for risk.

14. Limitations, Bias, and Responsible AI

The model has useful business value, but it also has limitations. The dataset is synthetic and relatively small, and it may not include all factors that influence real customer churn decisions. Real customers may leave or renew because of family needs, grocery prices, income, competitor offers, moving homes, delivery availability, product quality, or schedule changes.

Table 20: Limitations and mitigation plan

Limitation	Business Risk	Mitigation / Future Improvement
Synthetic and limited dataset	The data may not fully represent real customer behaviour.	Validate the model using real FreshBasket historical churn data before operational use.
Class imbalance	The No class is larger than the Yes class.	Evaluate recall, precision, F1-score, and confusion matrix, not accuracy alone.
False negatives still exist	Some actual churned customers may be missed.	Use the model as a screening tool and combine it with customer service knowledge.
Possible bias by region or customer group	Certain groups may receive different levels of attention.	Review predictions for fairness and avoid unfair or intrusive treatment.
Feature importance is not causation	The model shows useful splits but not direct causes.	Use business review, experiments, and customer feedback before changing policy.

Human judgment should still be used because the model cannot fully understand customer context, fairness concerns, privacy expectations, or the long-term customer relationship. Managers should review the model output and decide which retention action is appropriate and respectful.

15. Conclusion

This assignment demonstrates how a Decision Tree classification model can support customer churn prediction for FreshBasket. The project followed a structured machine learning workflow: business understanding, data inspection, duplicate removal, feature and target separation, train-test split, preprocessing, model training, evaluation, feature importance interpretation, overfitting review, and business recommendations.

The model achieved testing accuracy of 72.62% and recall for churned customers of 60.87%. The results suggest that the model can help FreshBasket identify many at-risk customers, but it should be used with human judgment because false positives, false negatives, synthetic-data limitations, and possible fairness concerns remain.

Appendix A: Assignment Alignment and Decision Tree Visualization

Table 21: Assignment alignment checklist

Assignment Question	Connection to FreshBasket
What is the business about?	FreshBasket is a subscription-based grocery delivery service.
What problem is the business trying to solve?	Customer churn and loss of recurring subscription revenue.
What decision can machine learning help make?	Which customers should receive proactive retention support.
What is the target variable?	Churned.
What metric is most important?	Recall for the Yes class is especially important because missing churned customers is costly.
Why use human judgment?	To review customer context, avoid unfair treatment, and choose appropriate retention actions.

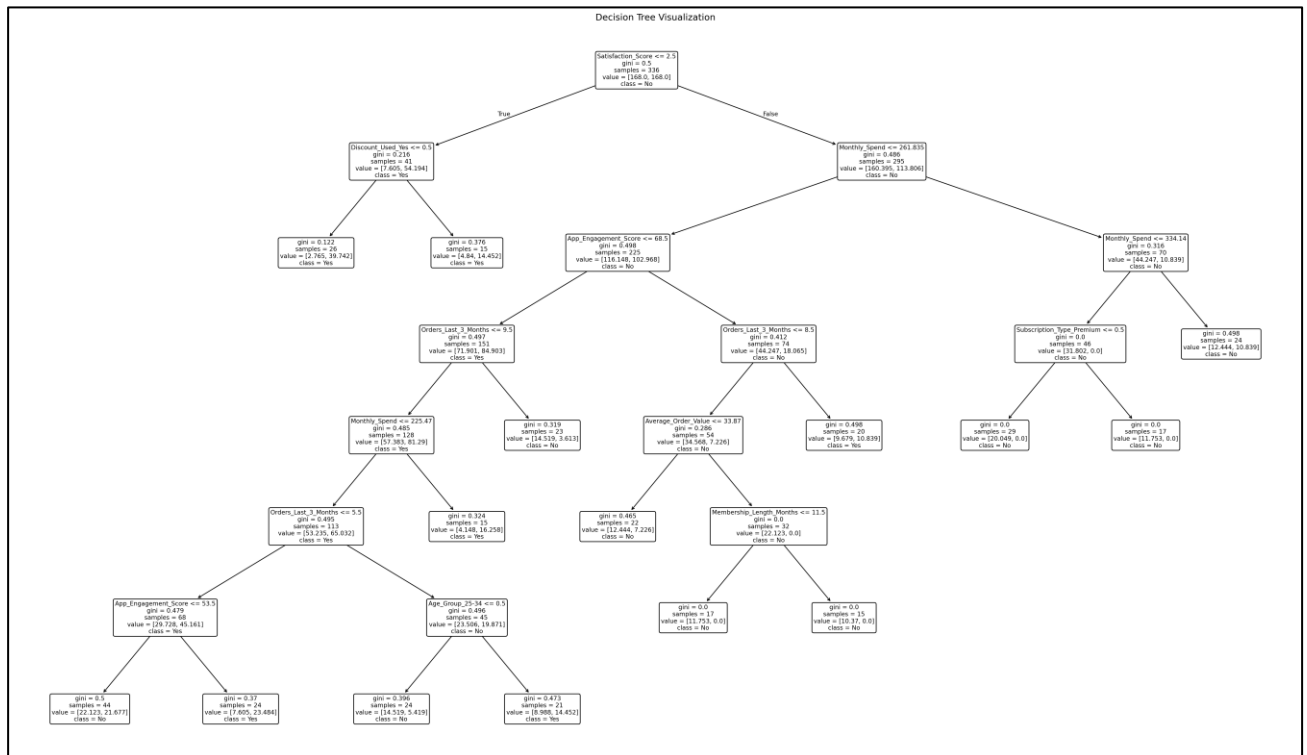


Figure 10: Decision Tree visualization